Surveys

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```
library(tidyverse)
theme_set(theme_bw())
```

1. Using the data from the shark BLL survey, calculate the mean abundance in each year SPHYRNA MOKARRAN (great hammerhead), assuming a simple random sample.

```
station<-read.csv("Station_Data.csv") %>%
  mutate(date=as.Date(DATE),
         year=year(date))%>%
  filter(year>2000)

catch<-read.csv("Catch_Data.csv")
```

Make a new column for species and count catch by species

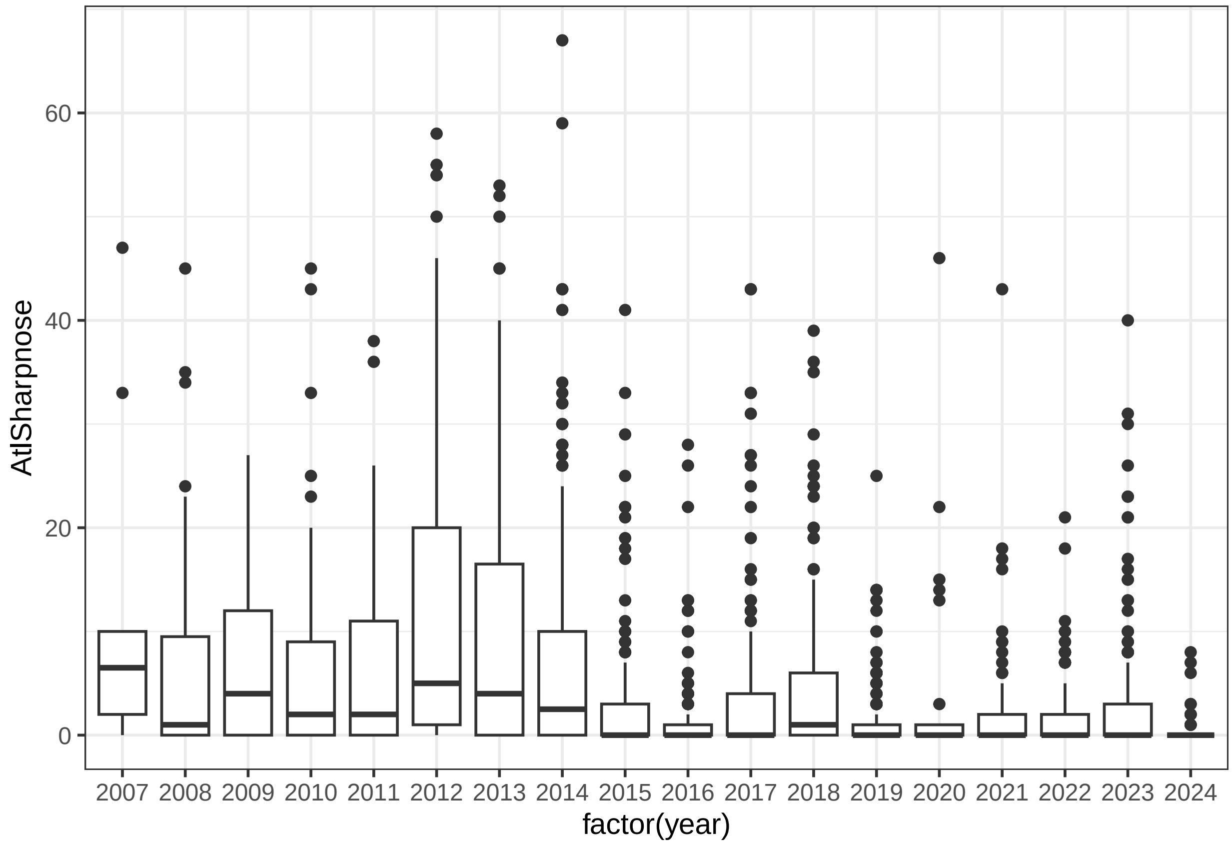
```
catchSum<-catch %>%
  mutate(Species=paste(GENUS,SPECIES)) %>%
  filter(Species %in% c("RHIZOPRIONODON TERRAENOVAE","SPHYRNA MOKARRAN"))%>%
  group_by(SEAMAPSTATION,CID,Species) %>%
  summarize(count=n())%>%
  pivot_wider(names_from=Species,values_from=count,values_fill = 0)
```

```
## `summarise()` has grouped output by 'SEAMAPSTATION', 'CID'. You can override
## using the `.groups` argument.
```

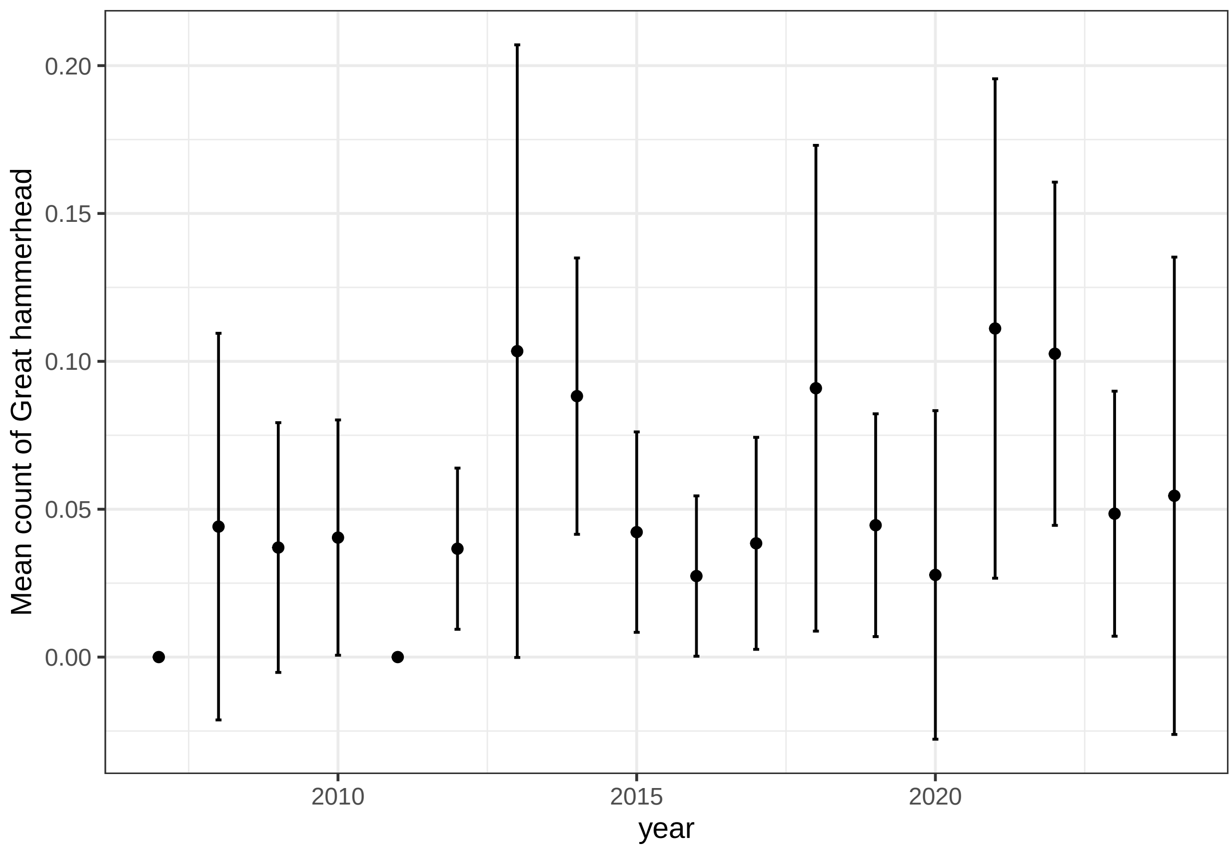
```
StationData<-left_join(station,catchSum,by=c("SEAMAPSTATION","CID"))
StationData<-StationData %>%
  mutate(greatHammer=replace_na(`SPHYRNA MOKARRAN`,0),
         AtlSharpnose=replace_na(`RHIZOPRIONODON TERRAENOVAE`,0))
```

Get annual mean numbers

```
ggplot(StationData,aes(x=factor(year),y=AtlSharpnose))+
  geom_boxplot()
```




```
yearSum<-StationData %>%
  group_by(year) %>%
  summarize(meanAS=mean(greatHammer),
            seAS=sd(greatHammer)/sqrt(n()))
ggplot(yearSum,aes(x=year,y=meanAS,ymin=meanAS-2*seAS,ymax=meanAS+2*seAS))+
  geom_point()+
  geom_errorbar(width=0.1)+
  ylab("Mean count of Great hammerhead")
```



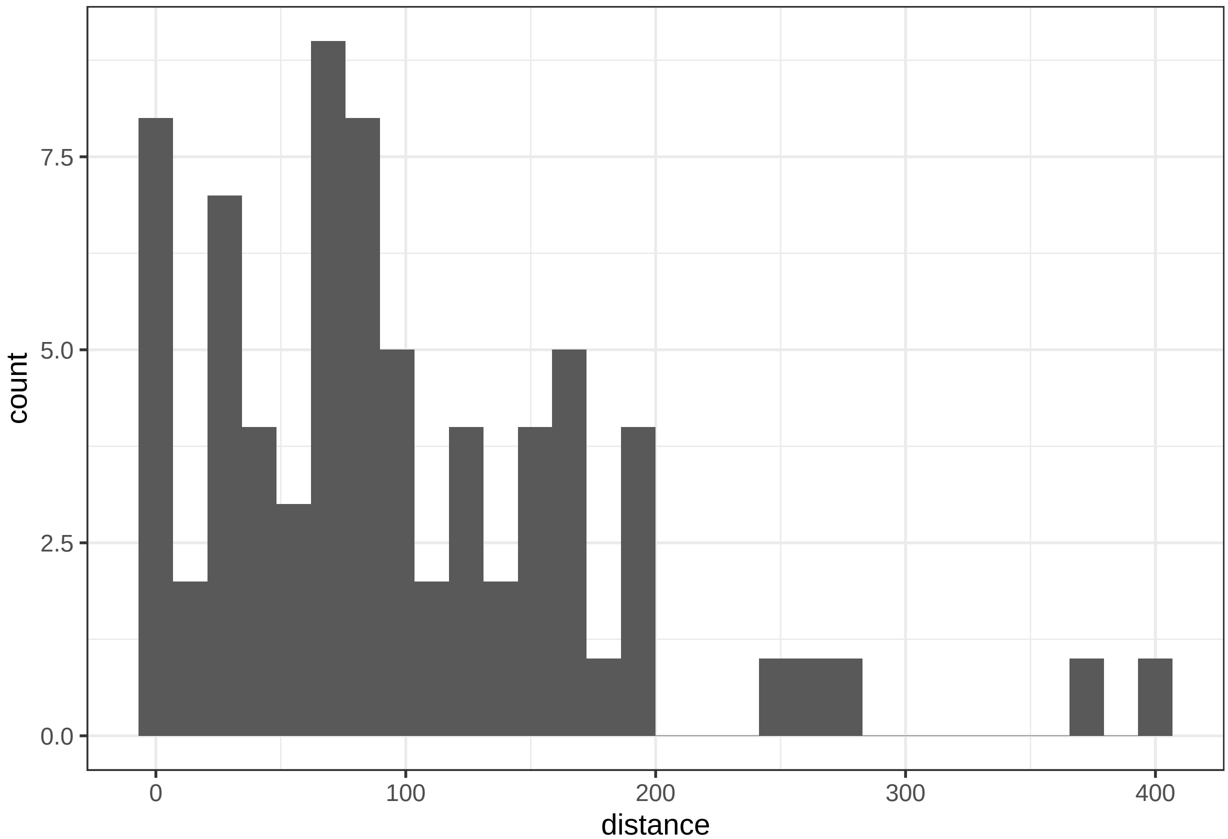
2. Using the impala data with distance sampling, use AIC to find the best functional form to model detection probability as a function of distance. Plot the detection functions. Does the AIC-best also look like the best fit? What is the total number in the sampled area and the density in the best model?

```
library(Distance)
```

```
## Loading required package: mrds

## This is mrds 3.0.1
## Built: R 4.5.1; ; 2025-10-26 02:26:06 UTC; windows

##
## Attaching package: 'Distance'

## The following object is masked from 'package:mrds':
##
##     create.bins
```

```
impala<-read.csv("impala.csv") %>%
  rename(distance=x)%>%  #column must be called distance (in meters)
  mutate(Effort=60000,
         Region.Label="combined",
         Sample.Label="combined")  #total length of transects in meters
ggplot(impala,aes(x=distance))+geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Both distance and total effort are in meters. We don't have information on individual transects or habitat area in the region, so we can only calculate density.

```
impalaHN<-ds(impala,
             key="hn",
             adjustment = NULL)
```

```
## No Area column found, generating density estimates only
```

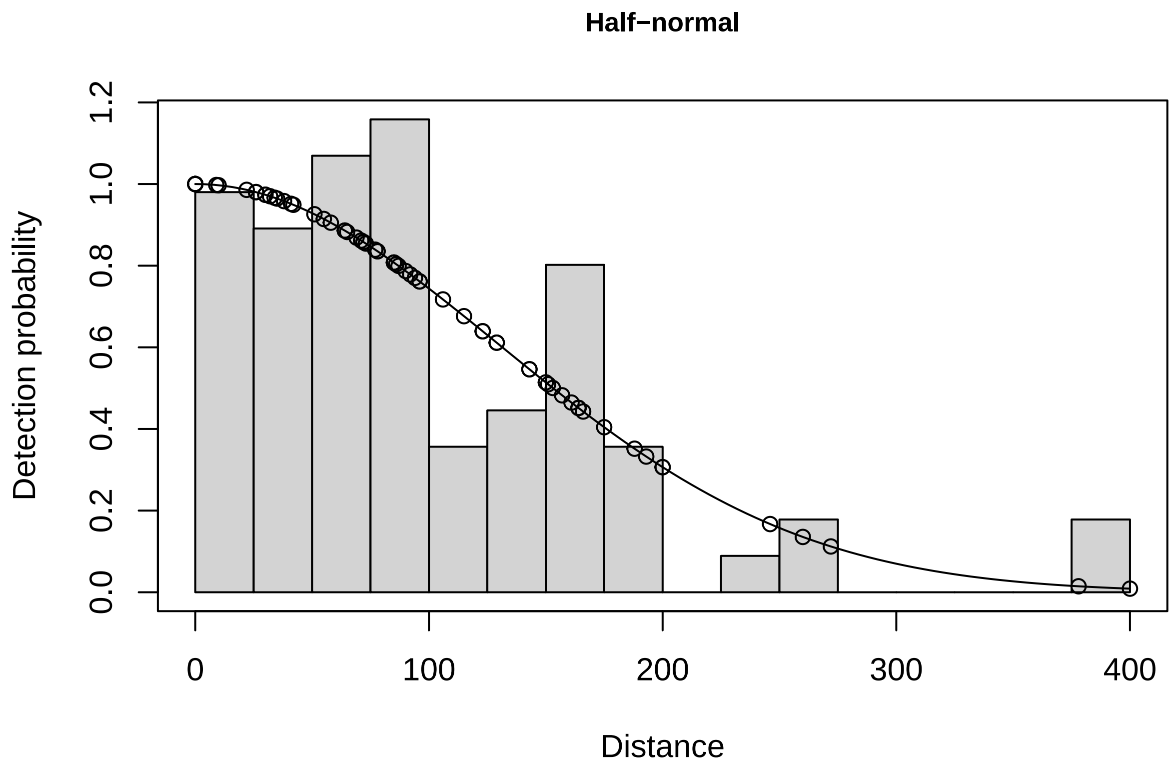
```
## Fitting half-normal key function
```

```
## AIC= 816.773
```

```
## Warning: Only one sample, assuming abundance in the covered region is Poisson.
## See help on dht.se for recommendations.
```

```
cutpoints=seq(0,450,25)
plot(impalaHN, breaks=cutpoints, main="Half-normal")
```

```
## Specified endpoints > 400; values reset.
```



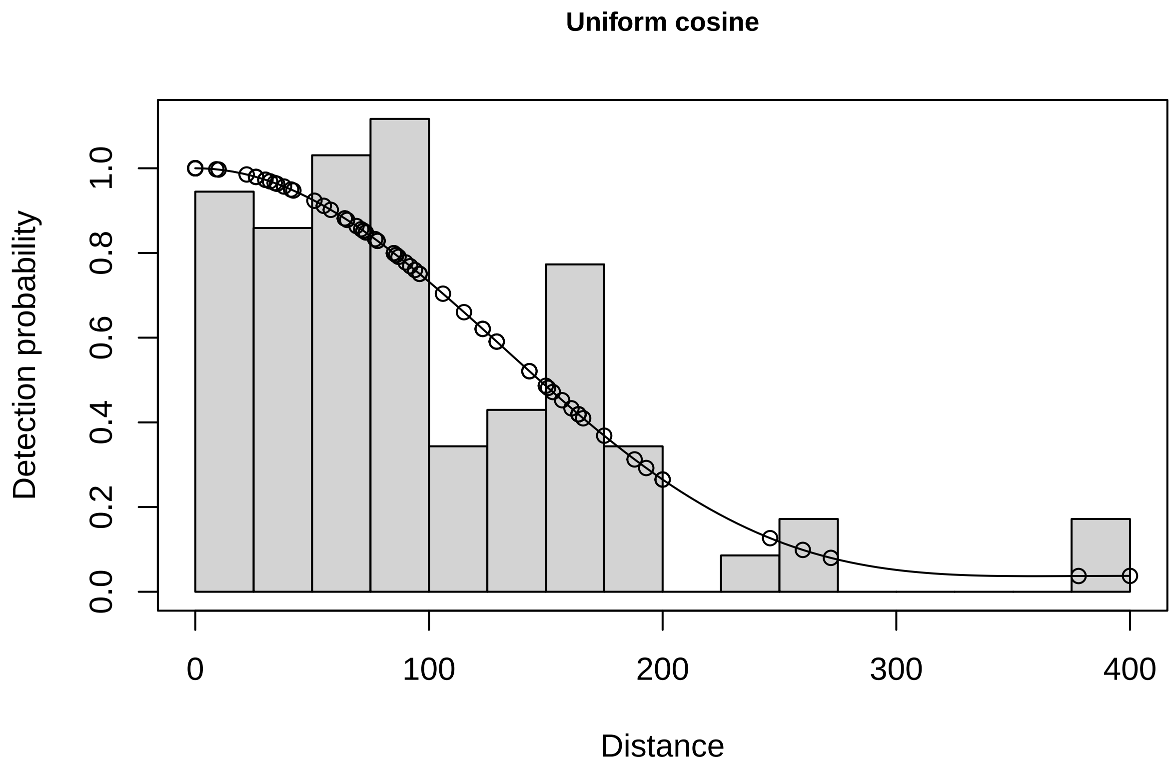

```
impalaUC<-ds(impala,
             key="unif",
             adjustment = "cos")
```

```
## No Area column found, generating density estimates only

## Starting AIC adjustment term selection.

## Fitting uniform key function

## AIC= 874.754

## Fitting uniform key function with cosine(1) adjustments

## AIC= 822.949

## Fitting uniform key function with cosine(1,2) adjustments

## AIC= 813.942

## Fitting uniform key function with cosine(1,2,3) adjustments

## AIC= 815.833
```

```
##
## Uniform key function with cosine(1,2) adjustments selected.

## Warning: Only one sample, assuming abundance in the covered region is Poisson.
## See help on dht.se for recommendations.
```

```
plot(impalaUC, breaks=cutpoints, main="Uniform cosine")
```

```
## Specified endpoints > 400; values reset.
```




```
impalaHP<-ds(impala,
             key="hr",
             adjustment = "poly")
```

```
## No Area column found, generating density estimates only

## Starting AIC adjustment term selection.

## Fitting hazard-rate key function

## AIC= 815.541

## Fitting hazard-rate key function with simple polynomial(4) adjustments
```

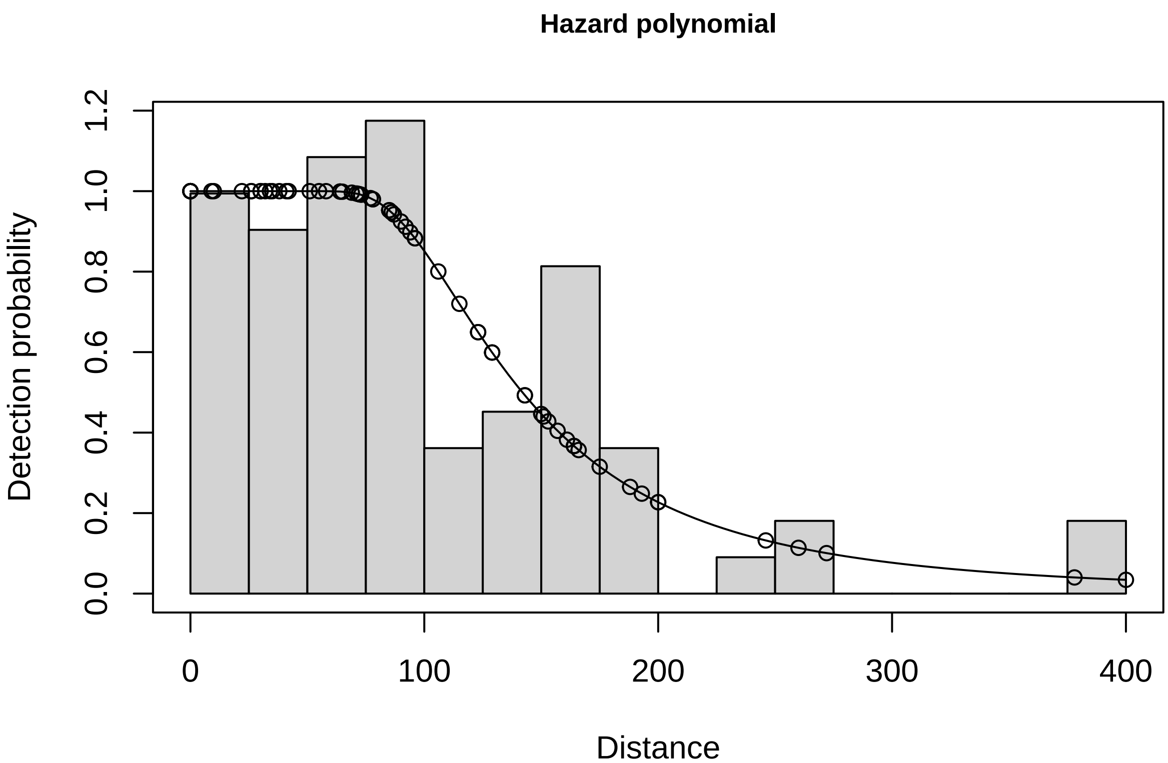
```
## AIC= 817.541
```

```
##
## Hazard-rate key function selected.
```

```
## Warning: Only one sample, assuming abundance in the covered region is Poisson.
## See help on dht.se for recommendations.
```

```r
plot(impalaHP, breaks=cutpoints, main="Hazard polynomial")
```

```
## Specified endpoints > 400; values reset.
```




```r
AIC(impalaHN,impalaUC,impalaHP)
```

```
##          df      AIC
## impalaHN  1 816.7735
## impalaUC  2 813.9418
## impalaHP  2 815.5413
```

uniform cosine is best

```r
summary(impalaUC)
```

```
##
## Summary for distance analysis
## Number of observations :  73
## Distance range         :  0  -  400
##
## Model       : Uniform key function with cosine adjustment terms of order 1,2
##
## Strict monotonicity constraints were enforced.
## AIC         :  813.9418
## Optimisation:  mrds (slsqp)
##
## Detection function parameters
## Scale coefficient(s):
## NULL
##
## Adjustment term coefficient(s):
##               estimate        se
## cos, order 1 1.2280310 0.1166683
## cos, order 2 0.3237394 0.1158605
##
##                         Estimate          SE         CV
## Average p              0.3918848  0.03416568 0.08718298
## N in covered region 186.2792404 23.51196571 0.12621893
##
## Summary statistics:
##     Region    Area CoveredArea Effort  n k          ER se.ER cv.ER
## 1 combined 4.8e+07     4.8e+07  60000 73 1 0.001216667     0     0
##
## Density:
##   Label     Estimate           se        cv          lcl          ucl       df
## 1 Total 3.880818e-06 4.419561e-07 0.1138822 2.928227e-06 5.143298e-06 5.676227
```